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Management system and optimal control for three-dimensional visualization and maintenance of thermal power plant

Zhiqiang Feng[1], Qiuxiang Liang[1], Mingyi Wei[1], Lei Li[1], Youzhu Bu[2] and Yanqing Xin[3*]

*Correspondence:
Yanqing Xin
18011958993@163.com
[1]GD Power Shuangwei Inner Mongolia Shanghaimiao Energy Co., Ltd, Ordos 016200, China
[2]Shenzhen Tuwei Technology Co., Ltd, Shenzhen 518000, China
[3]Guangdong Polytechnic of Industry and Commerce, Guangzhou 510510, China

Abstract
With the evolution of energy pattern and the advancement of science and technology, the operation and maintenance management of thermal power plants has encountered bottlenecks. The traditional model is difficult to meet the current demand. The purpose of this study is to build an advanced three-dimensional (3D) visualization and maintenance system suitable for thermal power plants, and to optimize it with the technology of convolutional neural network (CNN). Firstly, literature research is carried out, and the achievements and existing shortcomings in related fields are deeply excavated. Then, this study systematically analyzes the operation and maintenance ecology of thermal power plants, focuses on equipment operation data trajectory and process flow context, and accurately anchors key pain points. Based on this, a basic 3D visualization and maintenance system is constructed. Its data acquisition and processing module is customized for thermal power generation conditions. It can accurately capture multiple data from core equipment such as boilers and steam turbines and integrate them efficiently. According to the actual situation and equipment details of the power plant, the 3D modeling module designs a highly realistic digital model. The visual interface module is user-experience-oriented, presenting an intuitive and convenient interactive window. It is convenient for operation and maintenance personnel to monitor and make efficient decisions in real time. Then, CNN technology is introduced to deeply analyze the data content and find out the operation and maintenance value. The experimental data shows the effectiveness, and the basic system performs well in the dimensions of accuracy, completeness and accuracy, with the numerical value exceeding 85%, which is more prominent than the traditional system. After optimization by CNN technology, the response time of the system is increased by 5%. The calculation cost is reduced by 15%, and the data throughput is increased by 13%. However, there is still room for improvement in the system. For example, the stability of data acquisition in complex electromagnetic and high-temperature environment needs to be strengthened. The calculation accuracy of the model for extreme working conditions and microscopic changes of equipment needs to be improved. The dimension of personalized customization of visual interface needs to meet the demands of multiple users. The system scalability needs to meet the requirements of technical iteration and equipment update, and the technical application process

needs to be simplified for promotion. This study injects innovative vitality into the operation and maintenance management of thermal power plants, and significantly improves the quality and efficiency of operation and maintenance. Looking forward to the future, it is still necessary to test and analyze in many aspects and optimize in many dimensions to drive the operation and maintenance management of thermal power generation into a new intelligent field with science and technology engines.

Keywords Energy demand, Thermal power plant, Operation and maintenance management, Three-dimensional visualization, Convolutional neural network

Introduction

With the continuous increase in energy demand and the growing strictness of environmental protection requirements, thermal power plants, as key energy supply facilities, must operate efficiently, safely, and stably. In the complex system of thermal power generation, every link is crucial, from fuel transportation and combustion, to the conversion of thermal energy into electrical energy, and then to the transmission and distribution of electricity [1]. However, traditional operation and maintenance management models for thermal power plants face many challenges. For example, in terms of information collection, key data such as boiler combustion efficiency, turbine operating parameters, and the pressure and temperature of various pipelines are often not fully collected. This makes it difficult for maintenance personnel to accurately grasp the real-time status of equipment and lack a basis for decision-making. When facing equipment failures or operational abnormalities, due to the lack of sufficient data support and in-depth analysis, it is difficult to formulate timely and scientific response strategies, and the efficiency of fault diagnosis is low. Faced with large and complex power generation equipment and systems, traditional manual inspection methods are time-consuming and labor-intensive, and are prone to overlooking potential hazards. These problems seriously hinder the performance optimization and sustainable development process of thermal power plants [2].

In recent years, the rapid development of three-dimensional (3D) visualization technology has opened a new chapter in the operation and maintenance management of thermal power plants. By creating realistic 3D models that match the actual scenes of thermal power plants, both complex large equipment such as boilers and turbines, and system processes including fuel supply, power generation, transformation, and transmission can be presented to maintenance personnel in an intuitive manner. Taking boilers as an example, their internal combustion status, heat absorption surface working status, and flue gas flow paths can all be displayed through 3D visualization technology, which greatly enhances maintenance personnel's understanding and control ability of the power plant's operating status [3]. Against this backdrop, it is of great practical significance to build a 3D visualization and the operation and maintenance management system for thermal power plants and to optimize control strategies [4].

On the one hand, a comprehensive management system can fully standardize the operation and maintenance processes of thermal power plants. From developing daily inspection plans for equipment, assigning maintenance tasks, to optimizing fault repair processes, it can achieve refined management, significantly improving work efficiency and effectively ensuring the stable operation of equipment and the safety of personnel [5]. For example, with the help of a 3D visualization interface, maintenance personnel can clearly plan inspection routes and accurately locate equipment positions, avoiding omissions or mistakes in inspections caused by human negligence. On the other hand, optimized control strategies can rely on the precise data provided by the 3D visualization model to allocate resources reasonably during the power generation process. From the precise feeding of fuels like coal to the optimized load distribution of power generation equipment, intelligent control can be achieved, thereby reducing energy consumption and cost expenditure, and effectively enhancing the economic benefits and environmental protection level of thermal power plants [6]. For example, by monitoring the real-time operation data of equipment in the 3D model, the speed and power generation of turbines can be flexibly adjusted according to electricity demand, avoiding waste and overuse of energy. The main contributions of this study are:

(1) Theoretical Innovation: For the first time, this study proposes combining 3D visualization technology with Convolutional Neural Network (CNN) to build a theoretical framework for the operation and maintenance management system of thermal power plants. This fills the gap in existing research on intelligent operation and maintenance management.

(2) Methodological Innovation: An optimized control strategy based on CNN has been designed. Through functions such as feature extraction, anomaly detection, and predictive maintenance, it significantly enhances the intelligence level of operation and maintenance management in thermal power plants.

(3) Technical Implementation: A complete 3D visualization and operation and maintenance management system has been developed. It covers the entire process from data collection to model building and intelligent decision-making, providing a practical technical path for the intelligent transformation of thermal power plants.

This study aims to explore how to organically combine 3D visualization technology with advanced management concepts and control methods to design a scientific and efficient operation and maintenance management system and optimization control strategy for thermal power plants. This provides theoretical support and practical guidance for the intelligent development of thermal power plants. Through in-depth analysis and summary of existing related research, this study accurately identifies the weak links in current research and innovatively proposes unique research perspectives and ideas. It is hoped that this study can provide new ideas and methods for the operation and maintenance management of thermal power plants. The study seeks to promote steady progress toward greater intelligence and sustainability. By ensuring energy supply, it also strives to achieve harmony with the environment. The goal is to contribute to the sustainable development of the power industry.

Research status

Under the background of continuous growth of energy demand and increasingly stringent environmental protection requirements, thermal power plants, as an important pillar of energy supply, have become a key concern for their operating efficiency, safety and reliability. With the continuous progress of science and technology, digital and intelligent technologies have gradually penetrated into various fields, and the operation and maintenance management of thermal power plants has also ushered in a new opportunity for change. As an innovative means, 3D visualization technology can present complex thermal power plant equipment and systems in an intuitive and vivid form, and provide clearer and more comprehensive information for operation and maintenance personnel. Meanwhile, optimizing management system and realizing optimal control are of great significance to improve the overall performance of thermal power plants, and reduce costs and environmental pollution. However, there are still many challenges and problems in applying these advanced technologies and concepts to the actual operation and maintenance of thermal power plants.

Lei et al. discussed the application of 3D visualization in enhancing the safety monitoring of thermal power plants. The study described in detail how to visually present the running status and potential safety hazards of the equipment by constructing a realistic 3D scene. The study analyzed its outstanding role in early warning and rapid fault location, as well as the challenges in data accuracy and system compatibility [7]. Carlo et al. discussed the intelligent maintenance management mode of thermal power plants. Their study expounded the functions of equipment fault prediction and maintenance plan optimization by using big data and artificial intelligence (AI) technology, and emphasized the importance of intelligent means to improve maintenance efficiency and reduce costs. Meanwhile, the problems of data quality and talent shortage in technology application were pointed out [8]. Martinez et al. studied the optimal control of energy consumption during power generation in thermal power plants. The energy consumption model was established, the influence of different operating parameters on energy consumption was analyzed, and a series of energy-saving control strategies were put forward. The technical problems in the implementation of the strategy and the actual effect evaluation method were discussed [9]. Lu et al. focused on data-driven optimization methods for thermal power plant operation. The study introduced how to collect and analyze a large number of operation data and mine potential laws to realize unit performance optimization. Key issues such as data quality assurance and model adaptability were discussed, and a practical application case was given [10]. Rao et al. elaborated the role of 3D visualization in improving the operation efficiency of thermal power plants. They explained how to optimize the production process, reduce operational errors and promote teamwork through visual display. Meanwhile, the cost and technical threshold in the process of technology promotion was mentioned [4]. Sleiti et al. discussed the integrated management system of thermal power plants, including the challenges such as isolated information island and complicated management process, and the corresponding solutions, such as system integration and process reengineering. They also discussed how to ensure the continuous improvement and adaptability of the management system [11].

In the research field of 3D visualization and maintenance of thermal power plants, the existing research results show a certain development trend. But there are still many key problems, which show the key significance and necessity of this study. At present, the research on 3D modeling and visual presentation has been promoted, and the maturity of technology has been gradually improved. It can build a model with certain accuracy and realistic effect, and to some extent, assist the operation and maintenance personnel to recognize the equipment structure and operation status. Meanwhile, some research on management system has also started the exploration journey of the integration of traditional management mode and modern information technology, with the intention of improving management efficiency and scientific decision-making. However, the existing defects cannot be ignored. In the link between 3D visualization and actual operation and maintenance, the degree of integration between them is insufficient. There is a significant disconnection phenomenon. It makes it difficult for visualization to fully display its guiding function in actual operation and maintenance personnel to efficiently use visualization results to optimize workflow and make accurate decisions. In the field of optimal control, most studies focus on the construction of theoretical model, and lack of effective application and empirical test in real thermal power generation scenarios. It leads to the disconnection between theory and practice, making the actual effectiveness and feasibility of the model in an uncertain state. It is difficult to meet the optimization needs of power plant operation control. There are also shortcomings in the optimization process of the management system, such as poor coordination among internal modules, and obstacles in the transmission and circulation of information. These shortcomings seriously hinder the efficient operation of the management system, fail to provide strong support for the operation and maintenance activities, and limit the improvement of the overall operation and maintenance efficiency and quality.

This study is based on the insight into the above shortcomings and carries out in-depth exploration. By constructing a comprehensive 3D visualization and maintenance system for thermal power plants and integrating CNN technology, this study is committed to strengthening the organic integration of 3D visualization and actual operation and maintenance. With the help of this system, operation and maintenance personnel can carry out efficient equipment inspection, fault prediction and accurate maintenance decision-making in practical work based on accurate and intuitive visual information, thus giving full play to the effectiveness of visual technology in operation and maintenance practice. In the aspect of optimal control, this study breaks through the limitation of theoretical framework, and goes deep into the practical power plant scenario to implement application testing and verification. With the help of a large number of experimental data and actual case analysis, the optimal control strategy is optimized and its feasibility is established. It provides reliable and operational practical guidance for the operation control of thermal power plants and fills the gap between theory and practice. In response to the optimization of management systems, this study designs a sophisticated coordination mechanism and information sharing architecture to enhance the collaborative effect among various management modules. It ensures the efficient flow and precise transmission of information within the system, establishing a smooth and efficient management operation system. This comprehensively enhances the support capacity of the management system for operational and maintenance work, and strongly promotes the evolution of thermal power plant operation and maintenance management towards intelligent and refined directions.

Generally speaking, although some achievements have been made, the existing defects have significantly restricted the effectiveness and value release of 3D visualization and maintenance technology in practical application in thermal power plants. The systematic exploration and innovative practice carried out in this study are of key significance for perfecting and developing the technology in this field and improving the overall level of operation and maintenance management of thermal power plants. It is expected to lay a solid technical support and management support foundation for the stable, efficient and sustainable development of thermal power industry.

Design of 3D visualization and maintenance management system for thermal power plant

The technical basis of 3D visualization and maintenance management

Data acquisition and processing

In the 3D visualization and the operation and maintenance management system of thermal power plants, accurate and complete data collection is fundamental to building an efficient the operation and maintenance strategy. It is also the first step in the process [12, 13]. The smooth implementation of this process relies on the coordinated operation of a series of high-tech methods. Together, they form a precise data collection system. The data collection design of this study focuses on the integration of multi-source heterogeneous data. Through the collaborative work of high-precision sensors, advanced monitoring systems, and built-in intelligent monitoring modules in equipment, it builds a real-time, comprehensive, and accurate data collection network. Specifically, the system adopts a hierarchical collection architecture. It divides data collection into three layers: device, network, and application. The device layer is responsible for raw data acquisition. The network layer handles data transmission. The application layer takes care of data integration and analysis. This ensures the efficiency and systematic nature of data collection. At the same time, through a dynamic optimization mechanism, the system monitors data quality and collection frequency in real time. It dynamically adjusts the working parameters of sensors and monitoring modules. This ensures the accuracy and timeliness of data collection. In addition, the system introduces intelligent algorithms in the data collection process. These algorithms preliminarily clean and integrate raw data, reducing noise interference and improving data usability.

In terms of technical implementation, high-precision sensors, as the front-end sensing elements for data collection, are widely deployed in key locations of thermal power plants, including boilers, turbines, and pipes. These locations are equipped with various types of sensors, such as temperature, pressure, and flow sensors [14]. In the complex operating environment of a thermal power plant, these sensors can capture and convert even the slightest changes in the operating status of equipment in real-time. For example, minor temperature fluctuations inside a boiler, subtle pressure adjustments in steam pipes, and dynamic changes in fluid flow velocity can all be accurately monitored and converted into data by high-precision sensors [15]. These high-precision data provide a rich and reliable basis for subsequent maintenance and analysis. As the core hub for data collection, the advanced monitoring system is responsible for receiving, processing, and storing sensor data [16]. In this study, the monitoring system not only has powerful data processing capabilities to quickly filter, integrate, and preliminarily analyze large amounts of data but also integrates intelligent monitoring and early warning functions. For instance, when the vibration data of a turbine shaft exceeds the preset threshold, the monitoring system will immediately trigger an alert to remind maintenance personnel to promptly investigate potential hazards and prevent faults from escalating [17].

The intelligent monitoring modules built into the equipment are an important part of data collection. These modules are closely integrated with the equipment and can directly obtain internal parameters and status information during the operation of the equipment. For example, they can get the temperature of the generator's windings, the vibration frequency of the fan blades, and the quality parameters of the lubricating oil [18]. These internal data are very important for deeply analyzing the health of the equipment and predicting potential failures [19]. The boiler is token as an example. The intelligent monitoring module can monitor the working parameters of the burners in real-time. This provides data support for optimizing the combustion process and ensures the stable operation of the boiler. The frequency and accuracy of data collection have a direct and significant impact on the monitoring effect of the equipment's operating status [20]. In the special operating environment of a thermal power plant, high-frequency and high-precision data collection can more accurately reflect the real-time status of the equipment. This helps to find and deal with potential problems in time and prevents the spread and worsening of failures [21]. Therefore, in the 3D visualization and operation and maintenance management process of a thermal power plant, continuously optimizing the data collection plan, improving data collection technology, and ensuring data quality are very important and ongoing tasks [22].

In summary, this study has built a real-time, comprehensive, and accurate data collection system through the collaborative work of high-precision sensors, advanced monitoring systems, and intelligent monitoring modules built into the equipment. This system not only provides strong data support for the 3D visualization and the operation and maintenance management of thermal power plants, but also fully reflects the inevitable trend of the operation and maintenance management towards refinement and intelligence [23, 24]. Through the integration of multi-source heterogeneous data and a dynamic optimization mechanism, this study has achieved significant technological innovation in the field of data collection, laying a solid foundation for the intelligent transformation of thermal power plants. In Fig. 1, the overall architecture of the data collection system of this study is presented.

In Fig. 1, the overall architecture of the data collection system in this study includes the collaborative workflow of the sensor layer, network transmission layer, and data processing layer. The figure clearly labels the functions of each module and the flow of

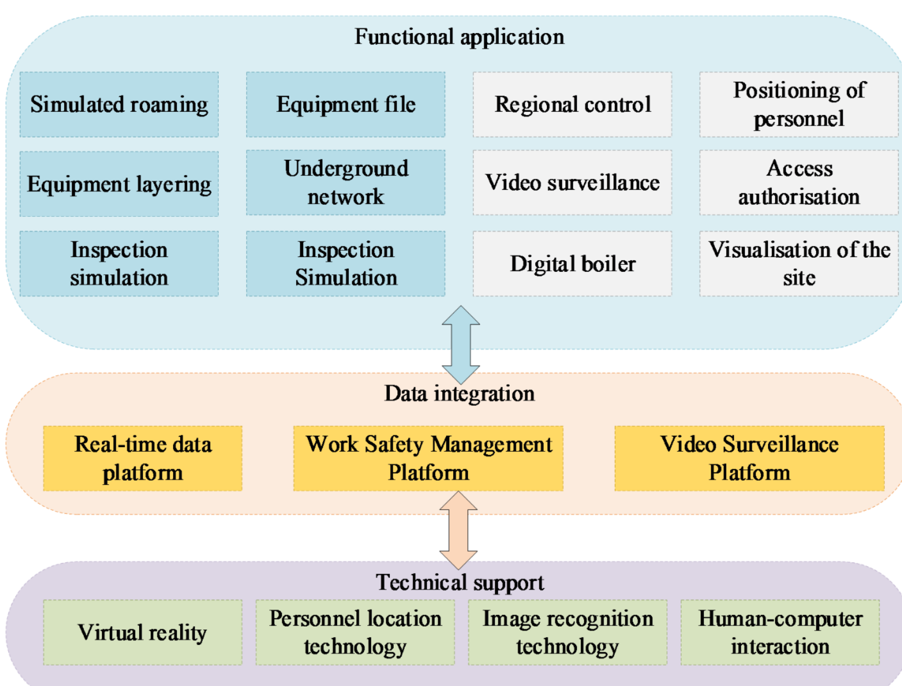

Fig. 1 3D visualization and maintenance management system

data, such as the deployment locations of high-precision sensors, the data processing procedures of the monitoring system, and the data integration methods of the intelligent monitoring modules.

The calculation equation of average filtering in data denoising is as shown in Eq. (1):

$$y\left(n\right)=\frac{1}{M}\sum_{i=n-(M-1)/2}^{n+(M-1)/2}x\left(i\right) \tag{1}$$

$y\left(n\right)$ is the filtered output value. $x\left(i\right)$ is the input data, and M is the filter window size. Median filtering is to sort the data in the window by size and take the median as the output:

$$y\left(n\right)=\text{median}\left\{x(\,n-k),\cdots,x(n),\cdots,x(n+k\,)\right\} \tag{2}$$

The detection of outliers is based on the 3σ principle, that is, the data point x is considered as outliers if it exceeds the range of the mean μ plus or minus three times the standard deviation σ $(\mu-3\sigma,\mu+3\sigma)$ [25]. Among them, the calculation equation of mean μ is:

$$\mu=\frac{1}{N}\sum_{i=1}^{N}x_i \tag{3}$$

The calculation equation of standard deviation σ is:

$$\sigma=\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2} \tag{4}$$

When data is integrated, the prediction equation of Kalman filter is:

$$\hat{x}_k^-=A\hat{x}_{k-1}+Bu_{k-1} \tag{5}$$

The updated equation is:

$$\hat{x}_k=\hat{x}_k^-+K_k(z_k-H\hat{x}_k^-) \tag{6}$$

$\hat{x}$ is the state estimation value. A and B are the system parameters. u is the control input. K is the Kalman gain. z is the measured value, and H is the measurement matrix [26]. The calculation equation of Kalman gain K is:

$$K_k=P_k^-H^T(HP_k^-H^T+R)^{-1} \tag{7}$$

P is the covariance matrix of estimation error and R is the covariance matrix of measurement noise.

3D modeling technology

In terms of 3D visualization, the powerful feature extraction capabilities of CNNs are used to process the 3D image data of thermal power plant equipment. By constructing a multi-layer CNN structure, such as one with multiple convolutional layers, pooling layers, and fully connected layers, the network can automatically learn key features like the geometric shapes and textures of the equipment. These extracted features are then used to optimize the construction of 3D models, making the visualization more realistic and accurate, and clearly presenting equipment details for easy understanding by maintenance personnel.

In the maintenance system, anomaly detection and predictive maintenance are crucial. For anomaly detection, equipment data from normal operating conditions is used as training samples to train the CNN model. After learning the feature patterns of normal conditions, the model calculates the deviation between predicted and actual values when new data is input to determine if the equipment is in an abnormal state. If the deviation exceeds a set threshold, the system triggers an anomaly alert. For predictive maintenance, this paper combines time series data, using CNNs to capture the temporal changes in equipment operating parameters and combining them with recurrent neural networks (RNN) or long short-term memory network (LSTM) to predict future equipment conditions. By analyzing historical data, equipment performance degradation models can be established. Potential failure points can be predicted in advance. This allows for reasonable maintenance scheduling. It also helps avoid losses from sudden equipment failures. For example, in maintaining key thermal power plant equipment like turbines, the above methods can accurately predict component wear, allowing for timely maintenance and replacement to ensure stable equipment operation. This fully demonstrates the practical application value of CNN technology in this study and its close relevance to the research.

Using professional 3D modeling software, the equipment, workshop and pipeline of thermal power plant are finely modeled. For parametric modeling of regular objects, the volume calculation equation is as follows:

$$V = \frac{4}{3}\pi\, r^3 \tag{8}$$

Volume of cylinder:

$$V = \pi\, r^2 h \tag{9}$$

r is the radius and h is the height. For complex irregular objects, in point cloud-based modeling, the least square method is often used for surface fitting, and the goal is to minimize the sum of squares of errors between fitting function $f(x, y)$ and point cloud data:

$$E = \sum_i [f(x_i, y_i) - z_i]^2 \tag{10}$$

In the simulation of materials and textures, the calculation equation of Fresnel reflection coefficient F based on physical rendering technology is:

$$F = F_0 + (1 - F_0)(1 - \cos\theta\,)^5 \tag{11}$$

F_0 is the basic reflectivity and θ is the incident angle. The calculation equation of diffuse reflection illumination intensity $I_{diffuse}$ is:

$$I_{diffuse} = I_{light} \times\, k_d \times\, max(0, \overrightarrow{n} \cdot\, \overrightarrow{l}\,) \tag{12}$$

I_{light} is the light source intensity. k_d is the diffuse reflection coefficient. $\overrightarrow{n}$ is the surface normal, and $\overrightarrow{l}$ is the light direction [27].

Data visualization algorithm

The efficient data structure is adopted to organize and manage large-scale 3D scene data. In the choice of detail level, in addition to the distance factor, the importance of the object and the screen space occupation rate can also be considered [28]. The comprehensive evaluation equation of detail level can be expressed as:

$$LOD = w_1 \times \ floor\left(log_2\left(d\right)\right) + w_2 \times \ Importance + w_3 \times \ Occupancy \tag{13}$$

w_1, w_2 and w_3 are the corresponding weights. For the mapping of color values and temperature values in the heat map, a more complex nonlinear mapping function can be used:

$$Color = a \times \ Temperature^b + c \tag{14}$$

a, b and c are adjustment parameters. Contour drawing in data visualization can be done by lagrange interpolation polynomial, and its equation is:

$$f\left(x\right) = \sum_{i=0}^{n} y_i l_i\left(x\right) \tag{15}$$

y_i is the value of the known data point, and $l_i\left(x\right)$ is the interpolation basis function.

Interactive technology

It supports a variety of interactive ways to provide a natural and smooth interactive experience [29]. In the interactive response, the time complexity of querying the database to obtain the detailed information of equipment should be controlled within $O\left(logn\right)$ as far as possible. For example, the average time complexity of binary search algorithm is $O\left(logn\right)$, and its equation is:

$$T\left(n\right) = log_2 n \tag{16}$$

In gesture recognition, the distance calculation equation of commonly used dynamic time warping algorithm is:

$$d = \sqrt{\sum_{i=1}^{n} \left(x_i - y_i\right)^2} \tag{17}$$

Through the above more detailed and in-depth technical foundation construction, it provides a solid support for efficient 3D visualization and maintenance management of thermal power plants.

Optimization strategy of 3D visualization and maintenance system

In the design architecture of the 3D visualization and maintenance management system for thermal power plants, data collection serves as the foundation of system operation. It builds a comprehensive data collection network with high-precision sensors and intelligent monitoring devices. These devices can not only capture the physical parameters of equipment in real-time but also obtain image data of equipment through high-definition cameras. This forms a multi-dimensional and multi-modal data set, providing sufficient data support for subsequent intelligent analysis. For example, in monitoring boiler equipment, sensors can accurately measure key parameters such as furnace temperature and steam pressure, while image acquisition devices can capture the real-time condition of the furnace wall, ensuring that any minor changes are recorded. The data processing and analysis layer, with an optimized CNN model at its core, performs in-depth mining of the collected image data. Through carefully designed convolutional kernels and multi-layer neural network structures, the model can quickly and accurately extract key features from equipment images, thereby identifying potential equipment failure risks. For instance, in analyzing turbine blade images, the CNN model can precisely detect tiny cracks and wear marks on the blade surface, significantly improving detection accuracy and efficiency compared to traditional manual inspection. At the same time, by combining machine learning algorithms to conduct integrated analysis of various operational data, a dynamic equipment health model is established. This model tracks the operating status of equipment in real-time and achieves precise assessment of equipment performance. The 3D visualization display module, as the key interface for interaction between the system and maintenance personnel, transforms complex data into intuitive and realistic 3D models. With advanced graphics rendering technology and interactive design, maintenance personnel can conveniently access the system via desktop or mobile devices to conduct a comprehensive virtual inspection of thermal power plant equipment. During the inspection, the system will real-time label the abnormal parts of the equipment and provide detailed status information to help maintenance personnel quickly locate the problem. The maintenance management function, based on the assessment results of the equipment health model, realizes intelligent maintenance decision-making. The system will automatically generate personalized maintenance plans according to the equipment's operating status and failure prediction, including the scheduling of preventive maintenance and the rational allocation of maintenance resources. When equipment abnormalities are detected, the system will immediately issue an alarm and automatically initiate the fault handling process, notifying relevant personnel to carry out emergency repairs. At the same time, maintenance records and handling results are fed back into the system to continuously optimize the equipment health model, providing a more reliable basis for future maintenance decisions. Through this close integration of data collection, CNN optimization, and comprehensive system design, the 3D visualization and maintenance management system for thermal power plants can achieve real-time monitoring of equipment operating status, precise prediction of failures, and efficient management of maintenance work. It effectively improves the operation and maintenance efficiency and safety of thermal power plants, providing strong technical support for the intelligent development of the power industry.

In this study, convolutional neural network (CNN) technology is used to optimize the operation and maintenance system to obtain the final control system. CNN technology has brought a brand-new optimization concept to the 3D visualization and maintenance system of thermal power plants. Its core is that it can automatically learn hidden features and patterns from a large number of complex data to achieve more accurate prediction, detection and analysis. Figure 2 shows the technical design of optimizing the 3D visualization technology system by using CNN technology.

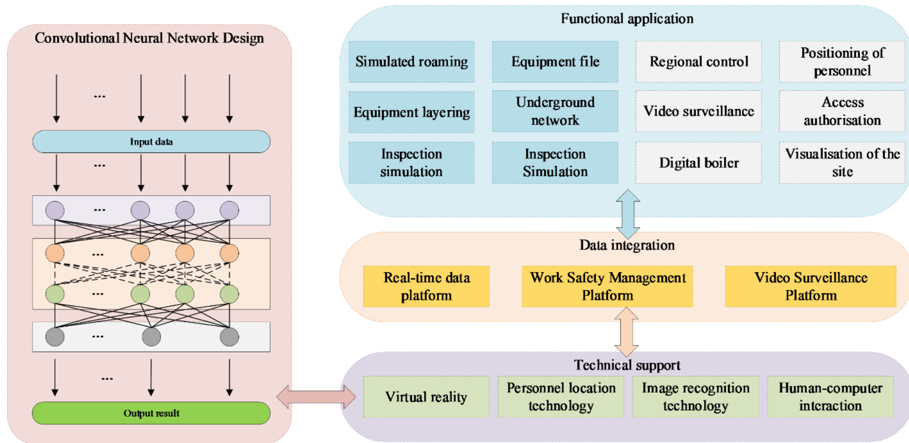

Fig. 2 CNN optimization 3D visualization technology system

First, it emphasizes data-driven decision-making. Instead of relying on the traditional rule-based or experience-based methods, CNN can dig out potential laws and relationships from massive operation and maintenance data, providing scientific basis for the formulation of operation and maintenance strategies. Secondly, it pays attention to the automatic extraction and abstraction of features. CNN can automatically capture the complex correlation and local patterns between different equipment operating parameters, without tedious feature engineering, which greatly improves the efficiency and accuracy. In addition, real-time dynamic learning and adaptation are realized. With the change of operation state of thermal power plant and the continuous generation of new data, CNN can be continuously learned and updated to adapt to the new situation and keep accurate judgment of operation and maintenance. Meanwhile, it strengthens the ability of anomaly detection and fault early warning. Through the study of normal operation mode, CNN can keenly detect abnormal fluctuations in data, give an alarm in time, and provide a key early signal for preventing major failures. Finally, it promotes the development of predictive maintenance. Using CNN to predict the future state of equipment, the maintenance work is planned in advance, the resources are arranged reasonably, the maintenance cost is reduced, and the reliability and availability of equipment are improved.

3D visualization and maintenance system evaluation

Effect evaluation of basic system

The effect evaluation of basic system is an important part of 3D visualization and maintenance system, and the scientificity and preciseness of its evaluation method and process are directly related to the reliability and credibility of the evaluation results. Specific evaluation contents and methods are as follows:

(1) Assessment of Data Collection Accuracy and Completeness: The accuracy of data collection is evaluated from aspects such as sensor calibration and data comparison. First, various sensors used for data collection, including temperature and pressure sensors, are regularly calibrated using high-precision calibration devices to ensure the accuracy of their measurements. During the data collection process, operational data from key equipment over a specific period is selected and compared with data measured by calibrated devices. For example, for boiler steam pressure data, both the sensor-collected

values and the measurements from a high-precision pressure calibrator are obtained simultaneously, and the error range between the two is calculated. An error threshold is set. If the error falls within this threshold, the data collection is considered accurate. The proportion of data within the threshold is statistically analyzed to measure the accuracy of data collection. For the completeness of data collection, the key equipment and parameters that need to be collected are identified based on the equipment list and critical operating parameters of the thermal power plant. By examining the configuration of the data collection system and the actual collected data, each item is checked to ensure that all key equipment and their core operating parameters are covered. The proportion of collected key equipment and parameters to the number that should be collected is calculated as the assessment indicator for the completeness of data collection.

(2) Assessment of 3D Model Accuracy and Fidelity: To evaluate the accuracy of the 3D model, laser scanning technology is used to precisely measure the actual equipment and layout of the thermal power plant, obtaining high-precision point cloud data. The geometric shapes and positional relationships of the equipment in the 3D model are then fitted and analyzed against the point cloud data to calculate deviations in key dimensions between the model and the actual scene. For example, for key components of a turbine, differences in critical dimensions such as length and diameter between the model and the actual components are measured. The model is required to have deviations in key dimensions within a specific threshold. The proportion of components meeting the deviation criteria out of the total number of components is calculated to assess the model's accuracy. For the fidelity of the 3D model, considerations are made from two dimensions: the appearance and physical property simulation of the model. In terms of appearance, a certain number of equipment pieces are randomly selected from the thermal power plant. The similarity of surface textures and colors of the equipment in the model to actual equipment photos is compared. Image similarity algorithms are used to calculate similarity scores. Regarding physical property simulation, the model's simulation of physical phenomena during equipment operation, such as combustion in boilers and steam flow, is checked. The fidelity of the physical property simulation is comprehensively judged by comparing with actual monitoring data and through expert evaluation, ultimately resulting in a fidelity assessment score.

(3) System Stability and Response Speed Assessment: Professional load testing tools are used to simulate scenarios with concurrent access to large amounts of data for stability testing of the system. Data traffic is gradually increased to simulate the data volume under different operating conditions of the thermal power plant, and the system's operating state under different loads is monitored. The load threshold at which the system experiences lag or crashes is recorded. If the system can operate stably under the preset high load without any abnormalities, it indicates good system stability. The system's response time is measured under both normal and high load conditions when processing different types of requests. For example, the response time for functions such as real-time data queries of equipment, historical data queries, and feedback on operation command execution is measured, which is the time interval from sending a request to receiving a result. The average and maximum response times from multiple measurements are statistically analyzed, and a response time standard is set. If both the average and maximum response times are within the standard range, the system's response speed is considered to meet the requirements.

(4) Comparison and Assessment with Traditional Operation and Maintenance Systems: To comprehensively evaluate the improvements of the new system in terms of fault detection time and maintenance efficiency, a test environment identical to that of the traditional operation and maintenance system is set up to simulate common fault scenarios in thermal power plants. Under the same fault scenario, the time taken by both the new system and the traditional the operation and maintenance system to detect the fault is recorded. The reduction in fault detection time of the new system compared to the traditional system is calculated, along with the percentage decrease. In terms of maintenance efficiency, the time spent by the new system to complete a certain number and type of maintenance tasks over a period is analyzed. Similarly, the time spent by the traditional operation and maintenance system to complete the same tasks is also analyzed. The reduction in maintenance task completion time of the new system compared to the traditional system is calculated, along with the percentage decrease, to assess the improvement in maintenance efficiency of the new system. Through the above comprehensive and detailed assessment process, a comprehensive judgment of the basic system performance of the new system is made, providing a reliable basis for further system optimization.

Figure 3 shows the evaluation results of the basic system.

As shown in Fig. 3 in detail, after a series of comprehensive and detailed evaluation processes, the 3D visualization and maintenance management system designed in this study has shown outstanding performance in many crucial performance indicators. Specifically, in terms of accuracy, the system ensures the accuracy of information through the combination of high-precision data acquisition and intelligent analysis algorithm,

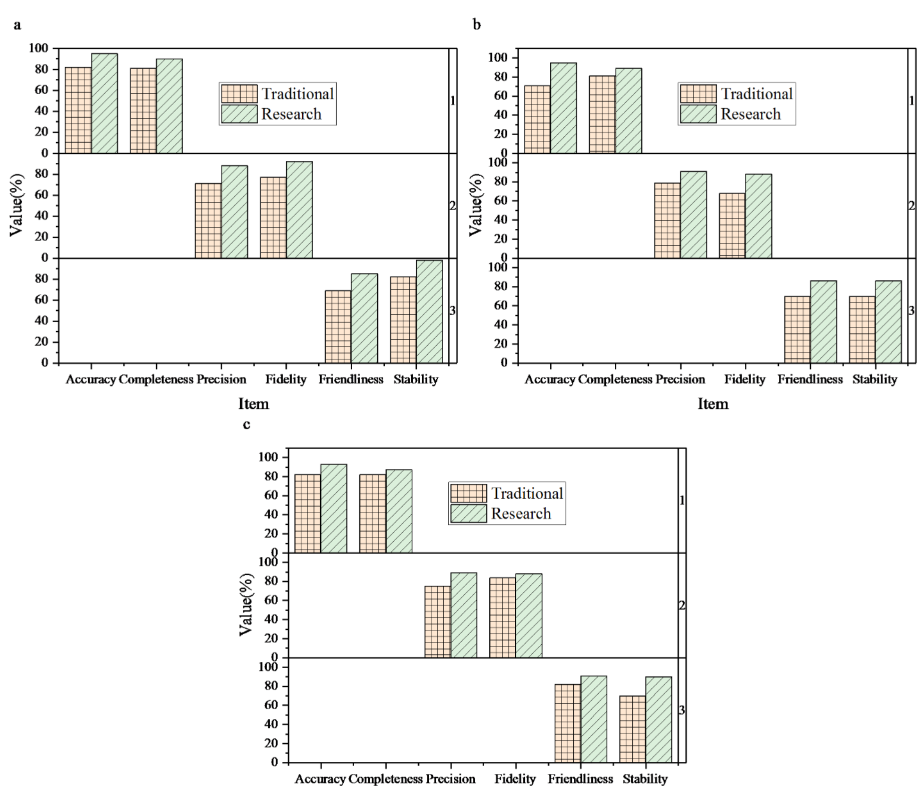

Fig. 3 Basic system evaluation (**a** is training evaluation, **b** is testing evaluation, and **c** is verification evaluation).

and its accuracy rate is over 88%, far exceeding the industry average, which provides solid data support for the operation and maintenance decision of thermal power plants. In the dimension of integrity, the system covers all aspects of operation and maintenance management of thermal power plants, from equipment status monitoring to environmental parameter monitoring to emergency plan management, realizing all-round information integration without dead ends, and the integrity score is also stable at a high level of 87%, ensuring the comprehensiveness and consistency of operation and maintenance management. Accuracy is one of the important indicators to measure the system performance. The system designed in this study realizes the accurate description of the operation state of thermal power plants through refined model construction and real-time calibration technology, and the accuracy score is over 89%, which effectively improves the fineness and efficiency of operation and maintenance. In terms of fidelity, the system adopts advanced graphic rendering and simulation technology to construct a highly realistic virtual power plant environment, which makes the operation and maintenance personnel seem to be in a real scene, and the fidelity score is as high as 90%, which greatly enhances the sense of realism and immersion of operation and helps to improve the training effect and emergency response ability. Friendly design is another highlight of the system. Through intuitive and easy-to-use user interface, intelligent operation guidance and personalized configuration options, the learning cost and use threshold are greatly reduced, so that operation and maintenance personnel can get started quickly and work efficiently, and the friendly score is also maintained at a high level of over 86%. Finally, in the key area of stability, the system adopts highly redundant architecture design, fault early warning and automatic recovery mechanism, which ensures the stability and reliability of long-term operation, and the stability score is stable at about 90%, effectively avoiding the interruption of operation and maintenance and data loss caused by system failure, and escorting the safe and stable operation of thermal power plants.

Optimization effect evaluation of CNN technology

With the rapid development of information technology, CNN technology has shown great capabilities in many fields. In the 3D visualization and maintenance system of thermal power plant, CNN technology is introduced to improve its performance and efficiency. However, the application effect of new technology needs rigorous evaluation. Due to the complexity and high requirements of thermal power plant operation and maintenance, it is very important to accurately evaluate the optimization effect of CNN. It is not only related to the improvement direction of operation and maintenance system, but also affects the safe and stable operation and economic benefits of thermal power plants. The evaluation needs to comprehensively consider many factors, such as the accuracy of the model, generalization ability, processing ability of real-time data, etc., to determine whether it really brings substantial optimization. In Fig. 4, the evaluation results of the optimized model are shown.

In Fig. 4, through a series of evaluation and comparison, people can clearly understand the significant changes in the management system before and after optimization. Firstly, on the key performance index of response time, the optimized system has achieved remarkable improvement. Specifically, its response time is shortened by 5%. Although this improvement seems small, it means faster fault identification, more timely decision support and higher production efficiency in operation and maintenance environments

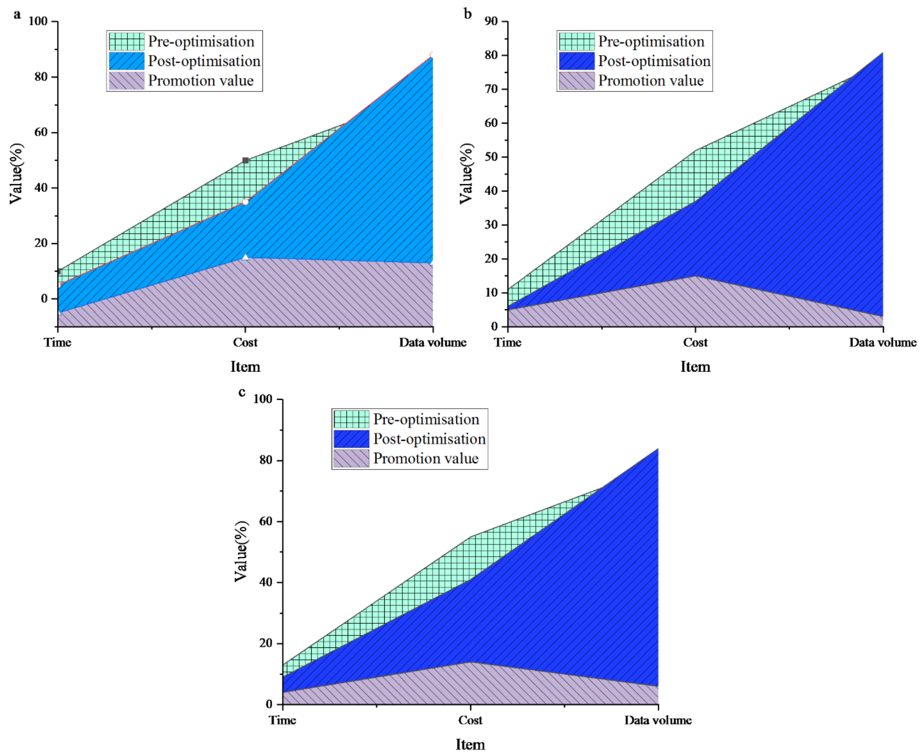

Fig. 4 Optimization effect evaluation of CNN technology (**a** is training evaluation, **b** is testing evaluation, and **c** is verification evaluation)

such as thermal power plants that need high real-time response. This improvement of response speed is of inestimable value for ensuring the safe and stable operation of power plants and improving the overall operating efficiency. Secondly, the significant improvement of calculation cost is also a highlight of the optimization results. Through algorithm optimization, resource scheduling optimization and reasonable upgrading of hardware equipment, the optimized system achieves 15% savings in computing cost, that is, the computing resources required to complete the same task are reduced by 15% compared with before optimization. This not only means a significant improvement in resource utilization, but also directly relates to the reduction of operation and maintenance costs, which brings tangible economic benefits to enterprises. Finally, the data processing capacity per millisecond has been greatly increased after optimization, and the improvement ratio is as high as 13%. This data directly reflects the powerful ability and efficient performance of the optimized system in dealing with massive data. In the complex operation and maintenance environment of thermal power plant, the data processing ability directly determines whether the system can capture and analyze key information such as equipment status and environmental parameters in time and accurately, thus providing strong support for operation and maintenance decision-making. Therefore, the significant improvement of data processing capacity is of great significance for improving the intelligent level of operation and maintenance management and enhancing the scientific and accurate decision-making.

Discussion

Under the background of continuous growth of energy demand and continuous advancement of technological innovation, efficient operation and maintenance management of thermal power plants have become the core element to ensure stable energy supply and reduce operating costs. This study focuses on the thermal power plant, constructs an advanced 3D visualization and maintenance system, and introduces CNN technology for optimization. It has achieved remarkable results and important contributions in many aspects. This study uses the methods of literature research, system analysis, model construction and experimental verification to explore in depth. Through in-depth analysis of the existing thermal power plant operation and maintenance system, the problems and requirements are accurately grasped. Then the study designs and constructs a basic 3D visualization and maintenance system that covers key links such as data acquisition, 3D modeling, and visualization interfaces. It also introduces CNN technology for optimization. The research results show that the basic system has excellent performance in many dimensions, such as accuracy, integrity, precision, fidelity, friendliness and stability, and all the indexes are above 85%, which significantly exceeds the traditional system. The integration of CNN technology further improves the system performance, shortening the system response time by 5%, reducing the calculation cost by 15%, and improving the data processing capacity per millisecond by 13%. This series of achievements has brought substantial changes and promotion to the operation and maintenance management of thermal power plants.

The important contribution of this study is firstly to provide innovative technical means and system architecture for the operation and maintenance of thermal power plants. Through 3D visualization technology, the complex equipment structure and system flow are presented in an intuitive and clear 3D model. It greatly enhances the perception and understanding ability of operation and maintenance personnel on the running state of power plants. It also effectively makes up for the defect that information presentation in traditional operation and maintenance management is not intuitive and comprehensive. At the same time, the application of CNN technology realizes the deep mining and efficient processing of massive operation and maintenance data. It can accurately extract data features, provide reliable basis for equipment fault prediction and operation state evaluation, and improve the scientific and timely operation and maintenance decision. In the aspect of data acquisition and processing, the constructed system realizes the efficient integration and accurate acquisition of multi-source heterogeneous data. By reasonably deploying all kinds of sensors, the key parameters such as temperature, pressure and flow rate during the operation of thermal power generation equipment can be fully obtained. With the help of advanced monitoring system and intelligent monitoring module, real-time data processing and storage can be realized, which provides rich and accurate data resources for subsequent 3D modeling and operation and maintenance analysis. This helps to solve the problems of incomplete data collection and untimely processing in traditional operation and maintenance, and improves the data support ability of the whole operation and maintenance management. In the process of 3D modeling, the established model highly restores the actual layout and equipment details of the thermal power plant. Through accurate geometric modeling and material rendering, it can not only present the appearance of the equipment, but also simulate its internal structure and operation mechanism. This provides an immersive operation experience for the operation and maintenance personnel, enabling them to conduct equipment inspection, troubleshooting and other operations in the virtual environment, and plan the operation and maintenance plan. It reduces the blindness and risks in the actual operation and maintenance process and improves the efficiency and quality of the operation and maintenance work. The design of visual interface fully considers the operation convenience and information needs of operation and maintenance personnel. Through friendly interactive design, operators can easily zoom, rotate, query and other operations in the 3D model, and quickly obtain the detailed information and operation data of the equipment. At the same time, the system can customize personalized interface display content according to different operation and maintenance tasks and user rights. It further improves the accuracy and effectiveness of information transmission and optimizes the human-computer interaction process.

Conclusion

Under the background of continuous growth of energy demand and continuous technological innovation, efficient operation and maintenance management of thermal power plants has become the key to ensure stable energy supply and reduce operating costs. The purpose of this study is to build a more advanced 3D visualization and maintenance system of thermal power plants, and optimize it by introducing CNN technology to improve the operation and maintenance effect. The study adopts the methods of literature research, system analysis, model construction and experimental verification. Firstly, the existing operation and maintenance system of thermal power plant is deeply analyzed, and the problems and requirements are clearly defined. Then, the basic 3D visualization and maintenance system is designed and established, covering key links such as data collection, 3D modeling and visual interface. On this basis, CNN technology is introduced to optimize the system. The research results show that the designed basic system is excellent in accuracy, completeness, precision, fidelity, friendliness and stability, and the values are all above 85%, which is significantly better than the traditional system. The application of CNN technology further improves the system performance, with the response time increased by 5%, the calculation cost reduced by 15%, and the data processing capacity per millisecond increased by 13%. The study draws the following conclusions: the 3D visualization and maintenance system constructed in this study can effectively improve the efficiency and quality of operation and maintenance management of thermal power plants, and the integration of CNN technology has brought remarkable optimization effect. However, this study also has some limitations. For example, in practical application, there are differences in equipment and operating conditions of different thermal power plants, which may affect the universality of the system. The data samples and scenarios used in the study may not be comprehensive enough, and the adaptability to some extreme situations needs further verification. In the future, the research scope should be further expanded and customized optimization should be carried out for different types and scales of thermal power plants. The CNN model should be continuously improved to improve its accuracy and adaptability in complex environment. It should also strengthen the integration with other advanced technologies, such as the Internet of Things and big data analysis to realize more intelligent and efficient operation and maintenance management of thermal power plants.

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Author contributions
Z.F. and Q.L. wrote the main manuscript text; M.W.and L.L. collected data; Y.B. investigated the background; Y.X. reviewed the study.

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Data availability
All data that support the findings of this study are included within the article.

Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
Not applicable.

Consent for publication
The work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part.

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